

N SANJAY RAO

AI ENGINEER — LLM APPLICATIONS & RAG SYSTEMS

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SUMMARY

AI engineer with 2+ years of hands-on LLM experience and an open-source LLM application on GitHub: an end-to-end RAG pipeline with embeddings, FAISS vector search, hybrid BM25 retrieval, cross-encoder reranking, and streaming API integration with multiple LLM providers. Day job: evaluating and prompt-engineering GenAI outputs on a Google workflow at Cognizant. Strong Python; building toward agentic LLM systems. Based in Hyderabad.

PROJECTS

Free RAG Chatbot — Document Q&A over PDF / DOCX / CSV • github.com/nsanjayrao/free-rag-chatbot
Python • **FAISS Vector Search** • **Embeddings (SentenceTransformers)** • **BM25** • **Streamlit** • **Gemini & Groq (Llama 3.3 70B) APIs**

- Built a complete RAG pipeline: multi-format document ingestion, recursive chunking (1,000 chars / 200 overlap), local MiniLM embeddings on CPU, and FAISS cosine-similarity vector search — runs entirely on free tiers with no paid infrastructure.
- Improved retrieval quality with hybrid dense + BM25 scoring, LLM-based query expansion, and cross-encoder reranking; every answer carries numbered citations back to its source chunks.
- Integrated two interchangeable LLM providers (Gemini, Groq — OpenAI-compatible API) with SSE token streaming using raw `urllib` — no SDK dependencies — with full HTTP error surfacing.
- Engineered content-hash index caching to skip re-embedding unchanged documents; persisted per-document chat history, PDF transcript export, and a responsive UI.

Customer Churn Prediction

Python • **scikit-learn** • **Classification**

- Trained Logistic Regression and Random Forest models on a telecom dataset with feature engineering; reached ~85% accuracy and identified the key drivers of churn.

Sentiment Analysis on Hotel Reviews

Python • **NLTK** • **NLP**

- Processed 10,000+ reviews with tokenization and sentiment scoring to surface service-quality trends and improvement areas.

EXPERIENCE

Senior Process Executive — GenAI Content Evaluation (L2) • Cognizant, Hyderabad June 2024 – Present

- Evaluate and quality-assure LLM-generated outputs on a Google GenAI content-evaluation workflow, verifying 150+ cases daily at 99%+ accuracy against evolving policy guidelines.
- Progressed through eight increasingly technical evaluation workflows; currently training in **Prompt Factory**, the programme's most technical prompt-engineering track.
- Mentor L1 analysts on evaluation accuracy and consistency; awarded "Cheers" recognition three consecutive months for top productivity and quality.

Inventory Management Trainee • Genesis Poweronics India Pvt Ltd, Hyderabad Feb 2024 – June 2024

- Rebuilt inventory stock reporting into interactive Power BI and Excel dashboards, cutting reporting time by 30%; coordinated parts sourcing with Ashok Leyland for genset installations.

SKILLS

LLM / RAG: RAG pipelines, embeddings & vector search (FAISS), hybrid retrieval (BM25), cross-encoder reranking, prompt engineering, LLM output evaluation, Gemini & OpenAI-compatible APIs, streaming (SSE)
Programming: Python (Pandas, NumPy, scikit-learn, NLTK), REST API integration, SQL, Git & GitHub, Streamlit
Data & Visualization: Power BI, Matplotlib, Excel (incl. VBA), EWOQ (Google evaluation tool)

EDUCATION

B.Tech, Electronics & Communication — Vignan's Institute of Information Technology, AP 2023 • CGPA 7.19